Emre Cecanpunar

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Selected Open Source Work

Mesa 2025 – Present

GPU drivers and compilers

- Develop and debug C/C++ code across Mesa shader compilers and Vulkan drivers.
- Upstreamed focused fixes in Intel's Jay shader compiler and Mesa's KosmicKrisp Vulkan-on-Metal driver, covering allocation lifetime and failed Metal compiler creation; diagnose failures with targeted Vulkan reproducers, Vulkan CTS, IR inspection, and sanitizer builds.
- Used drm-shim for standalone RADV/ACO experiments without physical AMD hardware and documented the results in technical notes.

Linux kernel 2025 – Present

Platform drivers and DRM

- Contribute focused C fixes across platform/x86 and DRM drivers, including hp-wmi and i915.
- Diagnose memory-safety, concurrency, undefined-behavior, and error-path issues with focused reproducers and sanitizer builds.

LLVM 2026 – Present

MC/ARM

- Submitted a fix to use label-time ISA state for Thumb function symbols when a later `.type` directive marks the function.

Apache NuttX 2026

SoC bring-up and upstream fixes

- Upstreamed initial NuttX support for the Rockchip RK3588S SoC and Orange Pi 5 Pro, booting through U-Boot to NSH with GICv3, timer, UART, and PSCI support; project announcement.
- Fixed two EDID parsing bugs: incorrect standard-timing decoding and use of the wrong chromaticity macro.

Other upstream work 2026

- Merged a heap-buffer-overflow fix in NVIDIA egl-wayland2 and submitted a use-after-free fix in Wayland.

Experience

Embedded Software Engineer Intern, Baykar Technologies Feb 2025 – Jun 2025

T3 Gemstone platform

- Initiated the Apache NuttX port to the TI AM67A SoC: brought up an Arm Cortex-R5F core, configured the MPU and caches, defined the memory map, and implemented the early boot path.
- Wrote linker scripts, GPIO drivers, and UART drivers; debugged NuttX ELF loading through Linux remoteproc, using `readelf` to inspect load addresses, sections, and vector and resource-table requirements.

Avionics Software Developer, Atmaca Rocket Team Sep 2022 – May 2023

University rocketry team · TÜBİTAK competition

- Developed redundant-controller avionics firmware and a parachute-deployment algorithm using Kalman-filtered multi-sensor altitude estimates; the avionics technical report received a full score from TÜBİTAK.

Projects

hpfand 2026 – Present

Fan-control daemon for HP Victus and Omen laptops

- Built a fail-safe Bash fan-control daemon around the hp-wmi hwmon interface, with CPU/GPU sensor aggregation, fan-stall protection, emergency cooling, and verified handoff back to firmware.

Education

B.Sc. Computer Science & Engineering, Akdeniz University Sep 2022 – Jun 2026